

Power BI Project – Data Transformation & Data Modeling

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This project demonstrates the complete data preparation and modeling workflow in Power BI using three source files: *List of Orders.csv*, *Order Details.csv*, and *Sales Target.csv*. The objective was to clean, transform, merge, and relate these datasets to enable accurate profit analysis and performance tracking against sales targets.

Data Preparation (Power Query)

Data Import

Imported all three source files into Power BI and opened the Power Query Editor for transformations.

Row Limiting

Retained the first 500 rows from *List of Orders* for simplicity using *Keep Top Rows*.

Reason: Easier handling and faster processing.

Data Type Correction

Order Date → Date

Amount → Fixed Decimal Number

Target → Fixed Decimal Number

Reason: Ensures accurate calculations and efficient performance.

	Order ID	\$ Amount	Profit	Quantity
1	B-25601	1,275.00	-1248	
2	B-25601	66.00	-12	
3	B-25601	8.00	-2	
4	B-25601	80.00	-36	
5	B-25602	168.00	-111	
6	B-25602	424.00	-272	
7	B-25602	2,612.00	1151	
8	B-25602	561.00	212	
9	B-25602	119.00	-5	
10	B-25603	1,353.00	-60	
11	B-25603	24.00	-30	
12	B-25603	193.00	-166	
13	B-25603	180.00	5	
14	B-25603	116.00	16	
15	B-25603	107.00	36	
16	B-25603	12.00	1	
17	B-25603	38.00	18	
18	B-25604	63.00	17	
19	B-25604	157.00	5	
20	B-25605	75.00	0	
21	B-25606	87.00	-4	
22	B-25607	50.00	15	
23	B-25608	1,364.00	-1864	
24	B-25608	476.00	0	
25	B-25608	257.00	23	
26	B-25608	856.00	385	
27	B-25609	485.00	29	
28	B-25609	25.00	-5	
29	B-25610	1,076.00	-38	
30	B-25610	107.00	-54	
31	B-25610	68.00	-55	

Text Formatting

Standardized *CustomerName* by applying *Capitalize Each Word*.

Reason: Improves readability and consistency.

Power Query Editor - Query Settings

Query: [Sales Target] - Table.TransformColumns([Transformed], Text.Combine([CustomerName], Text.Combine([State], Text.Combine([City], ""))))

Order ID	Order Date	Customer Name	State	City
1	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
2	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
3	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
4	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
5	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
6	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
7	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
8	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
9	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
10	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
11	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
12	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
13	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
14	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
15	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
16	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
17	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
18	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
19	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
20	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
21	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
22	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
23	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
24	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
25	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
26	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
27	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
28	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
29	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
30	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh
31	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh

Location Column
 Created a new *Location* field by merging *City* and *State* with a comma separator.
Reason: Simplifies mapping visuals.

Power Query Editor - Query Settings

Query: [Sales Target] - Table.TransformColumns([Table.TransformColumns([Transformed], Text.Combine([CustomerName], Text.Combine([State], Text.Combine([City], ""))), Text.Combine([State], Text.Combine([City], ""))])

Order ID	Order Date	Customer Name	State	City	Location
1	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
2	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
3	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
4	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
5	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
6	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
7	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
8	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
9	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
10	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
11	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
12	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
13	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
14	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
15	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
16	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
17	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
18	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
19	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
20	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
21	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
22	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
23	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
24	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
25	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
26	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
27	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
28	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
29	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
30	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh
31	01-01-2018	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh	Madhya Pradesh, Madhya Pradesh

New Columns & Calculations
Profit Margin
 Formula: Profit Margin = Profit / Amount
 Converted to percentage format.
Reason: Highlights profitability beyond revenue.

The screenshot shows a data table with the following columns: product, quantity, price, and profit. The data is organized into a grid with multiple rows and columns. The table is displayed in a software application window with a sidebar on the left and a top menu bar.

Profit Status

Conditional logic:

Profit < 0 → *Loss*

Profit = 0 → *Break-Even*

Profit > 0 → *Profit*

Reason: Enables quick filtering of profitable vs. non-profitable orders.

The screenshot shows a data table with the following columns: order ID, product, quantity, price, and profit. The data is organized into a grid with multiple rows and columns. The table is displayed in a software application window with a sidebar on the left and a top menu bar.

Data Integration & Validation

Merging Tables

Combined *List of Orders* and *Order Details* using *Order ID* with a Left Outer Join → *Orders Data*.

Reason: Consolidates order-level and product-level details.

Order ID	Order Date	Customer Name	State	City	Location	Order Date
9-25001	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25001
9-25002	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25002
9-25003	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25003
9-25004	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25004
9-25005	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25005
9-25006	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25006
9-25007	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25007
9-25008	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25008
9-25009	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25009
9-25010	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25010
9-25011	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25011
9-25012	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25012
9-25013	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25013
9-25014	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25014
9-25015	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25015
9-25016	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25016
9-25017	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25017
9-25018	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25018
9-25019	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25019
9-25020	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25020
9-25021	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25021
9-25022	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25022
9-25023	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25023
9-25024	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25024
9-25025	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25025
9-25026	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25026
9-25027	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25027
9-25028	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25028

Missing Data Check

Verified column quality → No missing values (due to row limit).

Duplicate Removal

Checked for duplicates → None found.

Sorting

Sorted *Order Date* in descending order.

Reason: Displays most recent sales first.

Filtering

Applied filter for *State = Tamil Nadu*.

Reason: Regional analysis practice.

Order ID	Order Date	Customer Name	State	City	Location	Order Date
9-25001	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25001
9-25002	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25002
9-25003	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25003
9-25004	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25004
9-25005	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25005
9-25006	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25006
9-25007	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25007
9-25008	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25008
9-25009	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25009
9-25010	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25010
9-25011	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25011
9-25012	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25012
9-25013	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25013
9-25014	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25014
9-25015	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25015
9-25016	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25016
9-25017	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25017
9-25018	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25018
9-25019	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25019
9-25020	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25020
9-25021	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25021
9-25022	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25022
9-25023	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25023
9-25024	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25024
9-25025	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25025
9-25026	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25026
9-25027	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25027
9-25028	25 April 2018	Shree	Tamil Nadu	Chennai	Chennai Tamil Nadu	9-25028

Aggregations & Grouping

Order Count

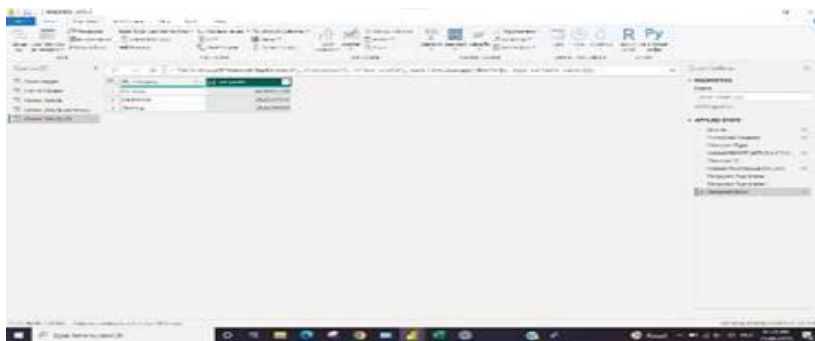
Grouped *Order Details* → Counted rows.

Outcome: Total orders placed.

Average Profit by Category

Grouped by *Category* → Calculated average profit.

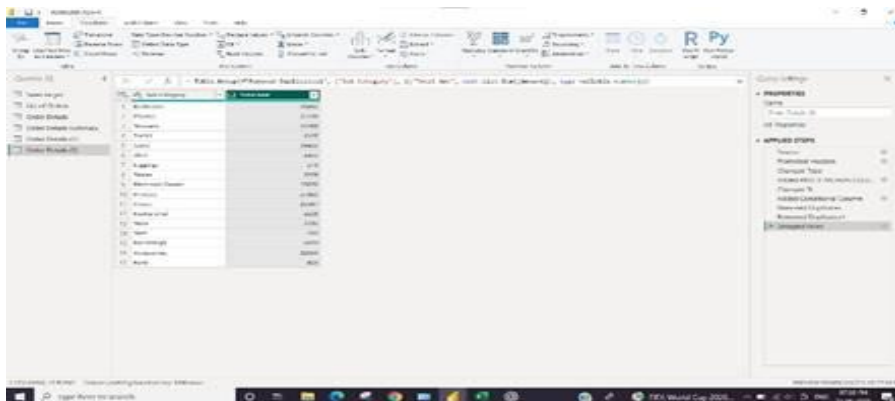
Outcome: furniture had the highest average profit (≈43.67).



Total Sales by Sub-Category

Grouped by *Sub-Category* → Summed *Amount*.

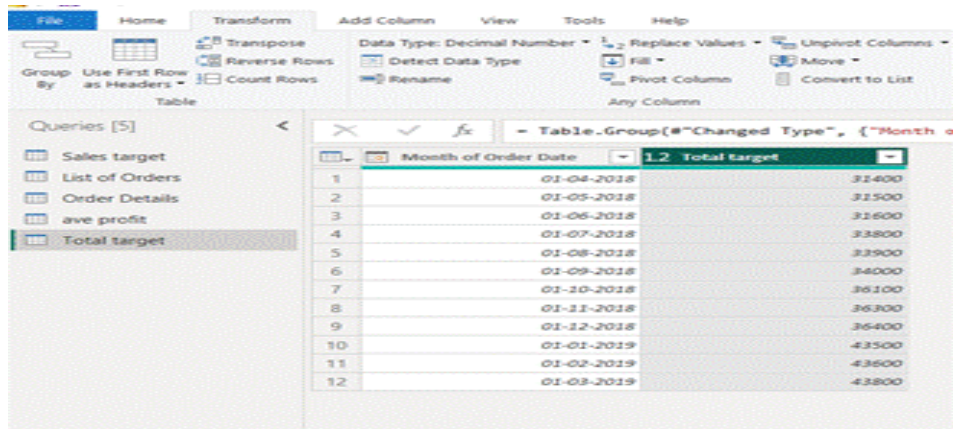
Outcome: Bookcase (₹30,841) and Printers (₹27,981) were top contributors.



Monthly Sales Targets

Grouped *Sales Target* by month → Summed *Target*.

Outcome: Targets ranged from ₹31,400 (Apr 2018) to ₹43,800(March 2019).

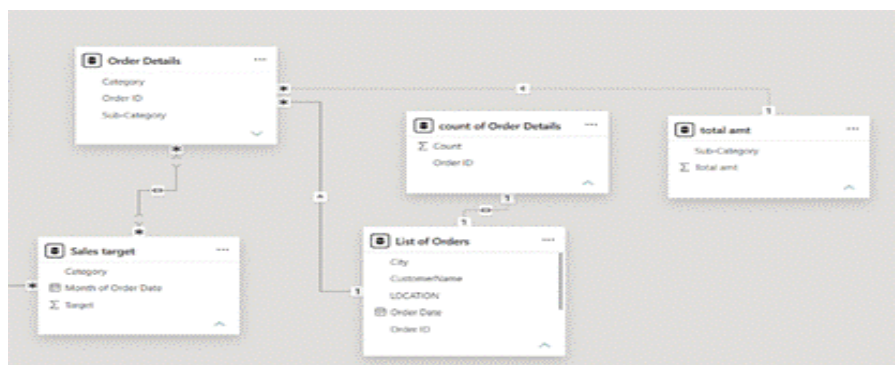


The screenshot shows the Microsoft Excel interface with a PivotTable. The PivotTable is titled "Table.Group(#'Changed Type', {'Month of Order Date', 'Total target'})". The PivotTable has two columns: "Month of Order Date" and "Total target". The data is grouped by month, showing the total target for each month from April 2018 to March 2019.

Month of Order Date	Total target
01-04-2018	31400
01-05-2018	31500
01-06-2018	31600
01-07-2018	31800
01-08-2018	31900
01-09-2018	34000
01-10-2018	36100
01-11-2018	36300
01-12-2018	36400
01-01-2019	43500
01-02-2019	43600
01-03-2019	43800

DATA MODELING

Use Manage Relationships



Conclusion

In this project, I imported raw data, cleaned up messy text, changed columns to the right formats, created formulas for profit, and connected the tables together. Now, the data is perfectly clean and ready to be used to build interactive charts and reports.

Thank You!